

git log/history

References

- [Visualizing Code: Polyglot Notebooks Repository \(YouTube\)](#)
- [gitstractor \(GitHub\)](#)
- [Neo4j Python Driver](#)

Command line execution (CLI mode): Yes

Git History - Directory Commit Statistics

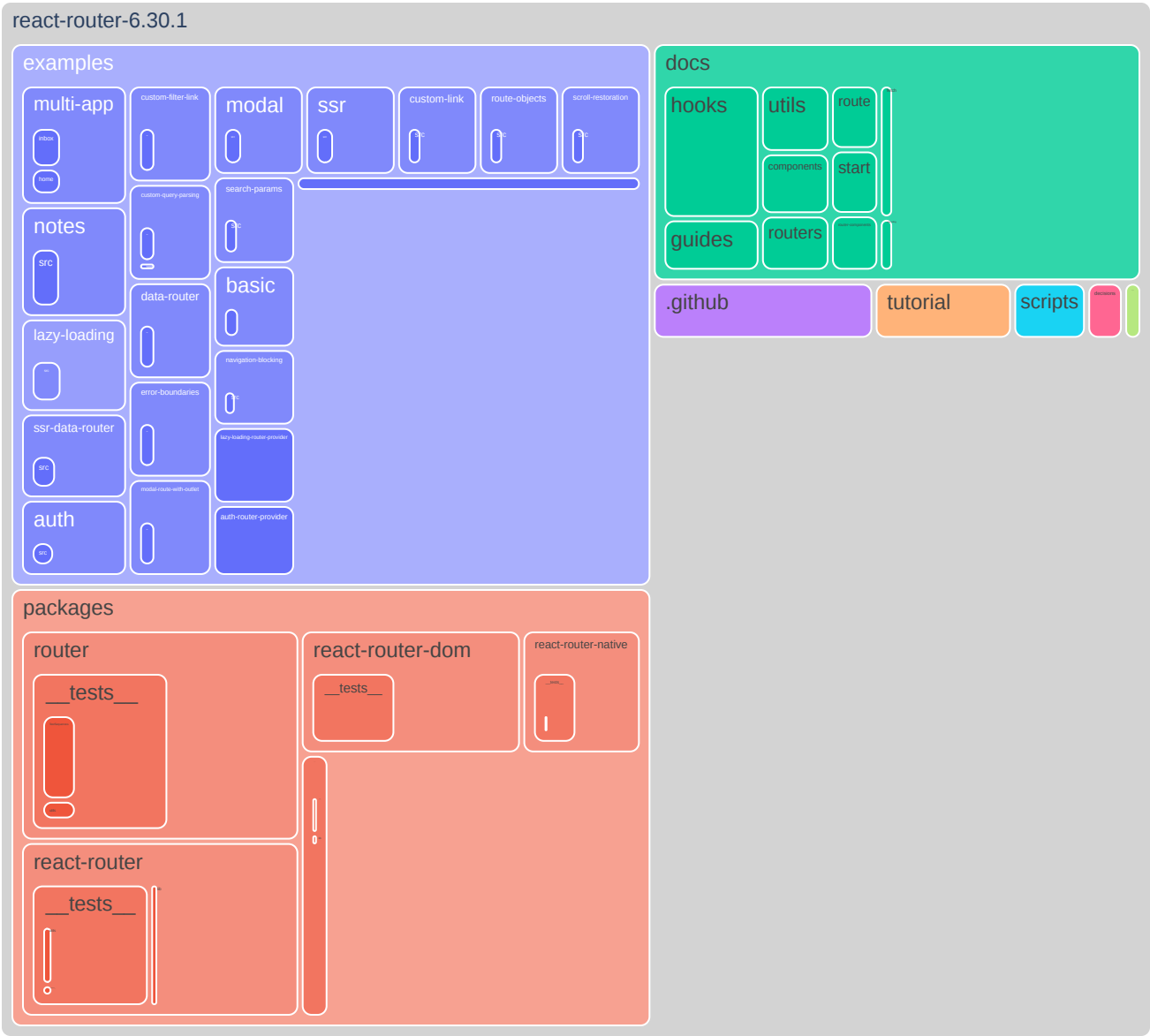
Data Preview

	fileCount	authorCount	commitCount	daysSinceLastCommit	daysSinceLastCreation	daysSinceLastModification
count	83.000000	83.000000	83.000000	83.000000	83.000000	83.000000
mean	25.313253	19.686747	130.457831	585.530120	999.626506	586.445783
std	77.867763	46.317549	293.197024	294.748025	389.268598	295.192465
min	1.000000	2.000000	3.000000	97.000000	117.000000	96.000000
25%	4.000000	4.000000	11.000000	327.000000	725.000000	326.000000
50%	10.000000	6.000000	26.000000	712.000000	1080.000000	711.000000
75%	13.000000	13.500000	84.000000	811.000000	1404.000000	810.000000
max	634.000000	344.000000	2000.000000	1445.000000	1501.000000	1444.000000

	directoryPath	directoryParentPath	directoryName	fileCount	firstFile	mostF
0	react-router-6.30.1/examples/custom-query-parsing/types	react-router-6.30.1/examples/custom-query-parsing	types	1	react-router-6.30.1/examples/custom-query-parsing/types/jsurl.d.ts	
1	react-router-6.30.1/packages/react-router-dom-v5-compat/lib	react-router-6.30.1/packages/react-router-dom-v5-compat	lib	1	react-router-6.30.1/packages/react-router-dom-v5-compat/lib/components.tsx	
2	react-router-6.30.1/packages/react-router-dom/__tests__/polyfills	react-router-6.30.1/packages/react-router-dom/__tests__	polyfills	1	react-router-6.30.1/packages/react-router-dom/__tests__/polyfills/drop-FormData-submitter.ts	
3	react-router-6.30.1/packages/react-router/__tests__/_snapshots__	react-router-6.30.1/packages/react-router/__tests__	__snapshots__	1	react-router-6.30.1/packages/react-router/__tests__/_snapshots__/route-matching-test.tsx.snap	
4	react-router-6.30.1/patches	react-router-6.30.1	patches	1	react-router-6.30.1/patches/@changesets_get-dependents-graph@1.3.6.patch	
5	react-router-6.30.1/.changeset	react-router-6.30.1	.changeset	2	react-router-6.30.1/.changeset/README.md	
6	react-router-6.30.1/docs/upgrading	react-router-6.30.1/docs	upgrading	2	react-router-6.30.1/docs/upgrading/future.md	
7	react-router-6.30.1/examples/lazy-loading/src/pages	react-router-6.30.1/examples/lazy-loading/src	pages	2	react-router-6.30.1/examples/lazy-loading/src/pages/About.tsx	
8	react-router-6.30.1/github/ISSUE_TEMPLATE	react-router-6.30.1/github	ISSUE_TEMPLATE	3	react-router-6.30.1/github/ISSUE_TEMPLATE/bug_report.yml	
9	react-router-6.30.1/examples/navigation-blocking/src	react-router-6.30.1/examples/navigation-blocking	src	3	react-router-6.30.1/examples/navigation-blocking/src/app.tsx	
10	react-router-6.30.1/packages/react-router-dom-v5-compat/__tests__	react-router-6.30.1/packages/react-router-dom-v5-compat	__tests__	3	react-router-6.30.1/packages/react-router-dom-v5-compat/__tests__/compat-router-test.tsx	
11	react-router-6.30.1/decisions	react-router-6.30.1	decisions	4	react-router-6.30.1/decisions/0001-use-blocker.md	
12	react-router-6.30.1/examples/basic/src	react-router-6.30.1/examples/basic	src	4	react-router-6.30.1/examples/basic/src/App.tsx	
13	react-router-6.30.1/examples/custom-link/src	react-router-6.30.1/examples/custom-link	src	4	react-router-6.30.1/examples/custom-link/src/App.tsx	
14	react-router-6.30.1/examples/custom-query-parsing/src	react-router-6.30.1/examples/custom-query-parsing	src	4	react-router-6.30.1/examples/custom-query-parsing/src/App.tsx	
15	react-router-6.30.1/examples/multi-app/home	react-router-6.30.1/examples/multi-app	home	4	react-router-6.30.1/examples/multi-app/home/App.jsx	
16	react-router-6.30.1/examples/route-objects/src	react-router-6.30.1/examples/route-objects	src	4	react-router-6.30.1/examples/route-objects/src/App.tsx	
17	react-router-6.30.1/examples/scroll-restoration/src	react-router-6.30.1/examples/scroll-restoration	src	4	react-router-6.30.1/examples/scroll-restoration/src/app.tsx	
18	react-router-6.30.1/examples/search-params/src	react-router-6.30.1/examples/search-params	src	4	react-router-6.30.1/examples/search-params/src/App.tsx	
19	react-router-6.30.1/packages/react-router-native/__tests__/_snapshots__	react-router-6.30.1/packages/react-router-native/__tests__	__snapshots__	4	react-router-6.30.1/packages/react-router-native/__tests__/_snapshots__/deep-linking-test.tsx.snap	
20	react-router-6.30.1/packages/react-router/lib	react-router-6.30.1/packages/react-router	lib	4	react-router-6.30.1/packages/react-router/lib/components.tsx	
21	react-router-6.30.1/packages/router/__tests__/_utils__	react-router-6.30.1/packages/router/__tests__	utils	4	react-router-6.30.1/packages/router/__tests__/_utils__/custom-matchers.ts	
22	react-router-6.30.1/tutorial/src	react-router-6.30.1/tutorial	src	4	react-router-6.30.1/tutorial/src/App.jsx	
23	react-router-6.30.1/docs/fetch	react-router-6.30.1/docs	fetch	5	react-router-6.30.1/docs/fetch/index.md	
24	react-router-6.30.1/examples/auth/src	react-router-6.30.1/examples/auth	src	5	react-router-6.30.1/examples/auth/src/App.tsx	
25	react-router-6.30.1/examples/custom-filter-link/src	react-router-6.30.1/examples/custom-filter-link	src	5	react-router-6.30.1/examples/custom-filter-link/src/App.tsx	
26	react-router-6.30.1/examples/data-router/src	react-router-6.30.1/examples/data-router	src	5	react-router-6.30.1/examples/data-router/src/app.tsx	
27	react-router-6.30.1/examples/error-boundaries/src	react-router-6.30.1/examples/error-boundaries	src	5	react-router-6.30.1/examples/error-boundaries/src/app.tsx	
28	react-router-6.30.1/examples/modal-route-with-outlet/src	react-router-6.30.1/examples/modal-route-with-outlet	src	5	react-router-6.30.1/examples/modal-route-with-outlet/src/App.tsx	
29	react-router-6.30.1/examples/modal/src	react-router-6.30.1/examples/modal	src	5	react-router-6.30.1/examples/modal/src/App.tsx	

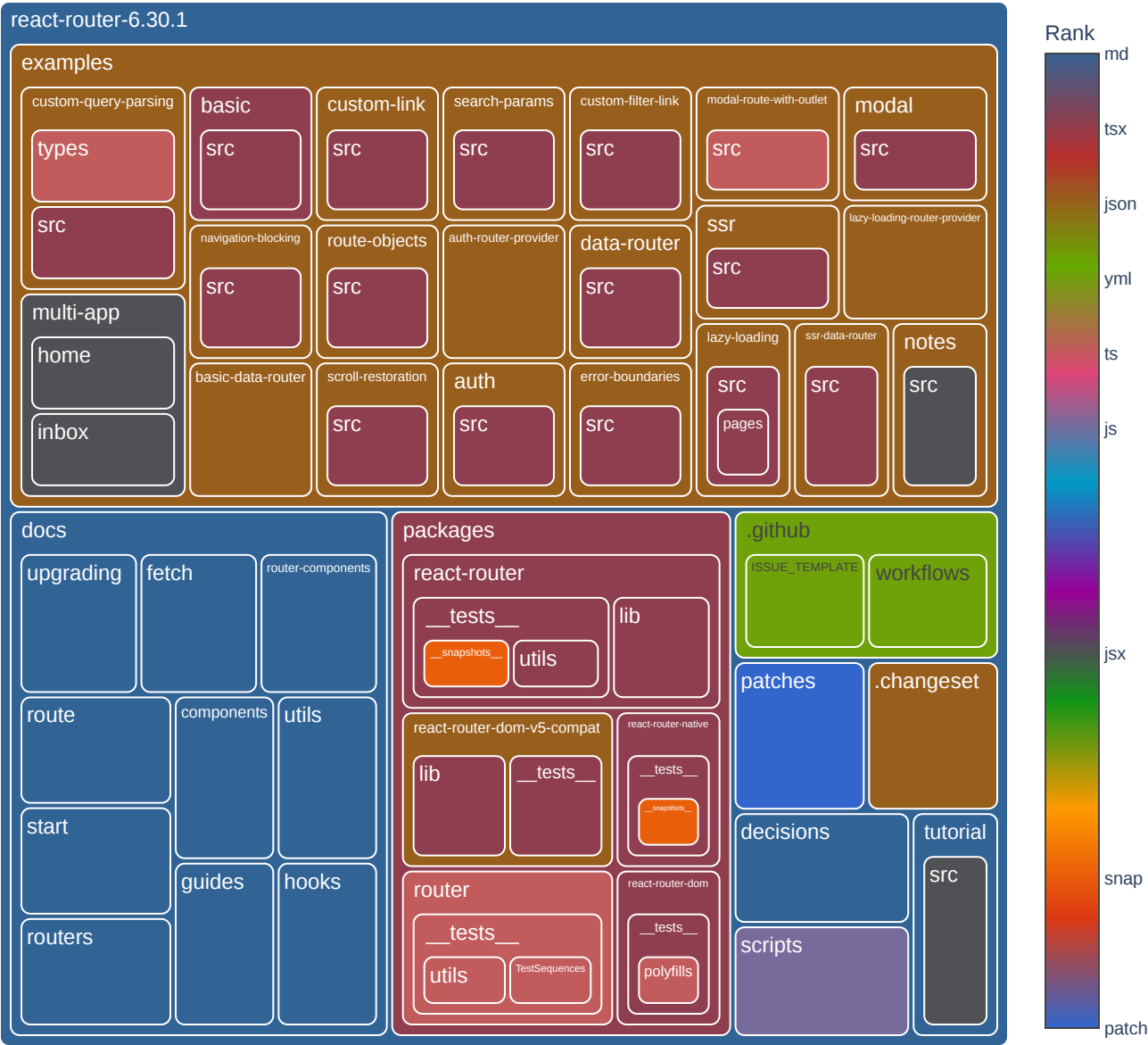
Number of files per directory

Directories and their file count



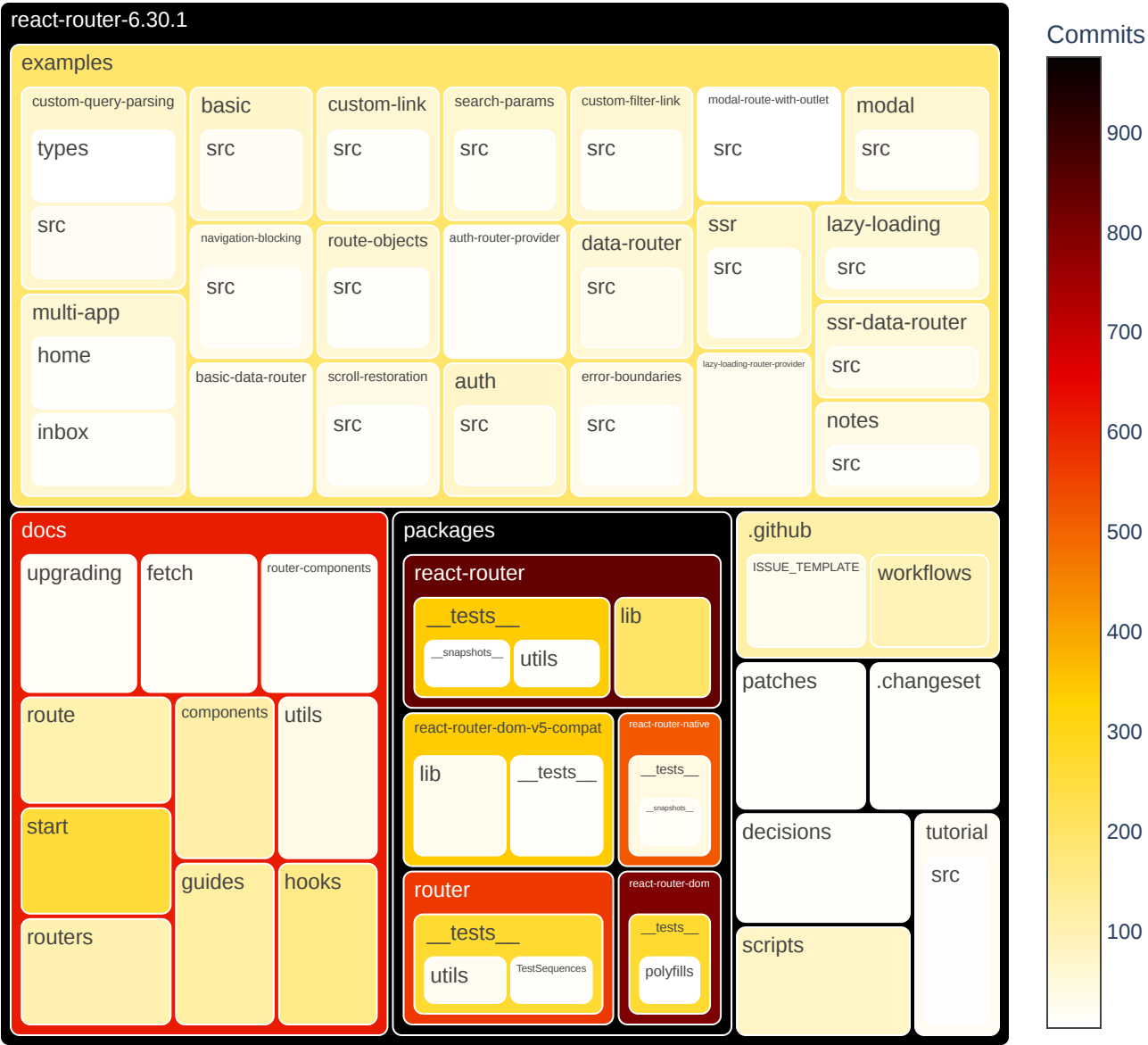
Most frequent file extension per directory

Most frequent file extension per directory



Number of commits per directory

Number of git commits



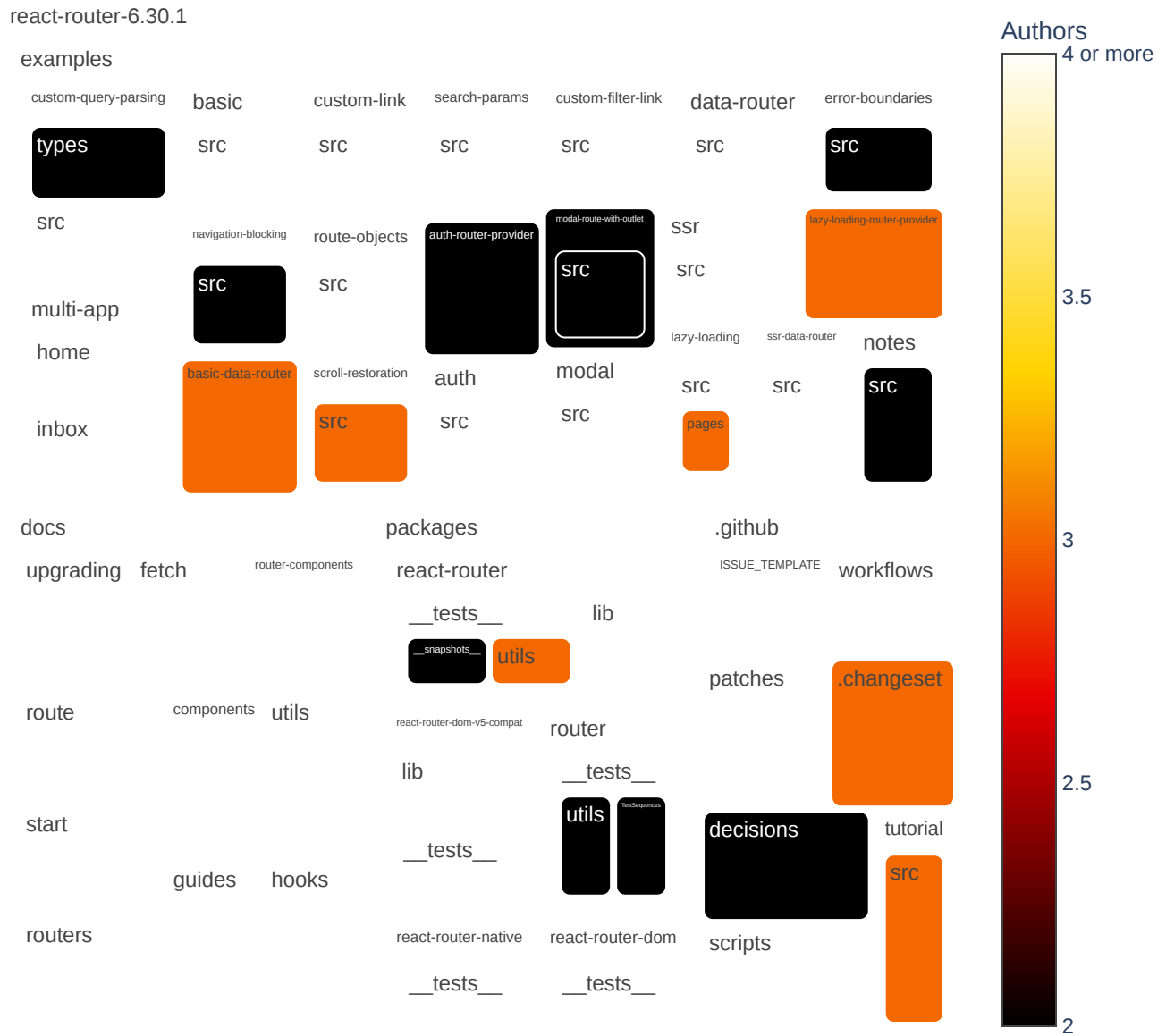
Number of distinct authors per directory

Number of distinct commit authors



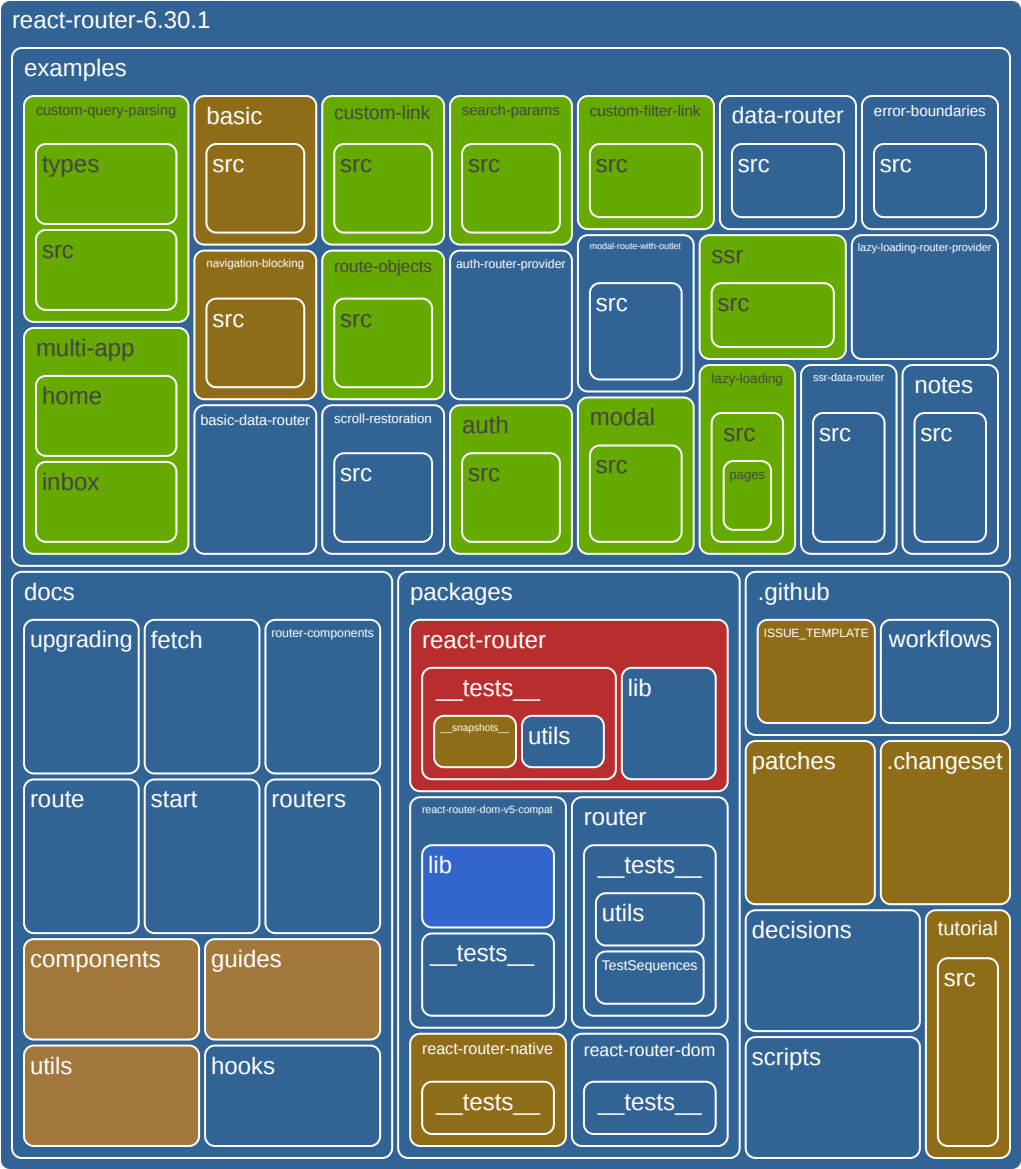
Directories with very few different authors

Number of distinct commit authors (red/black = only one or very few authors)



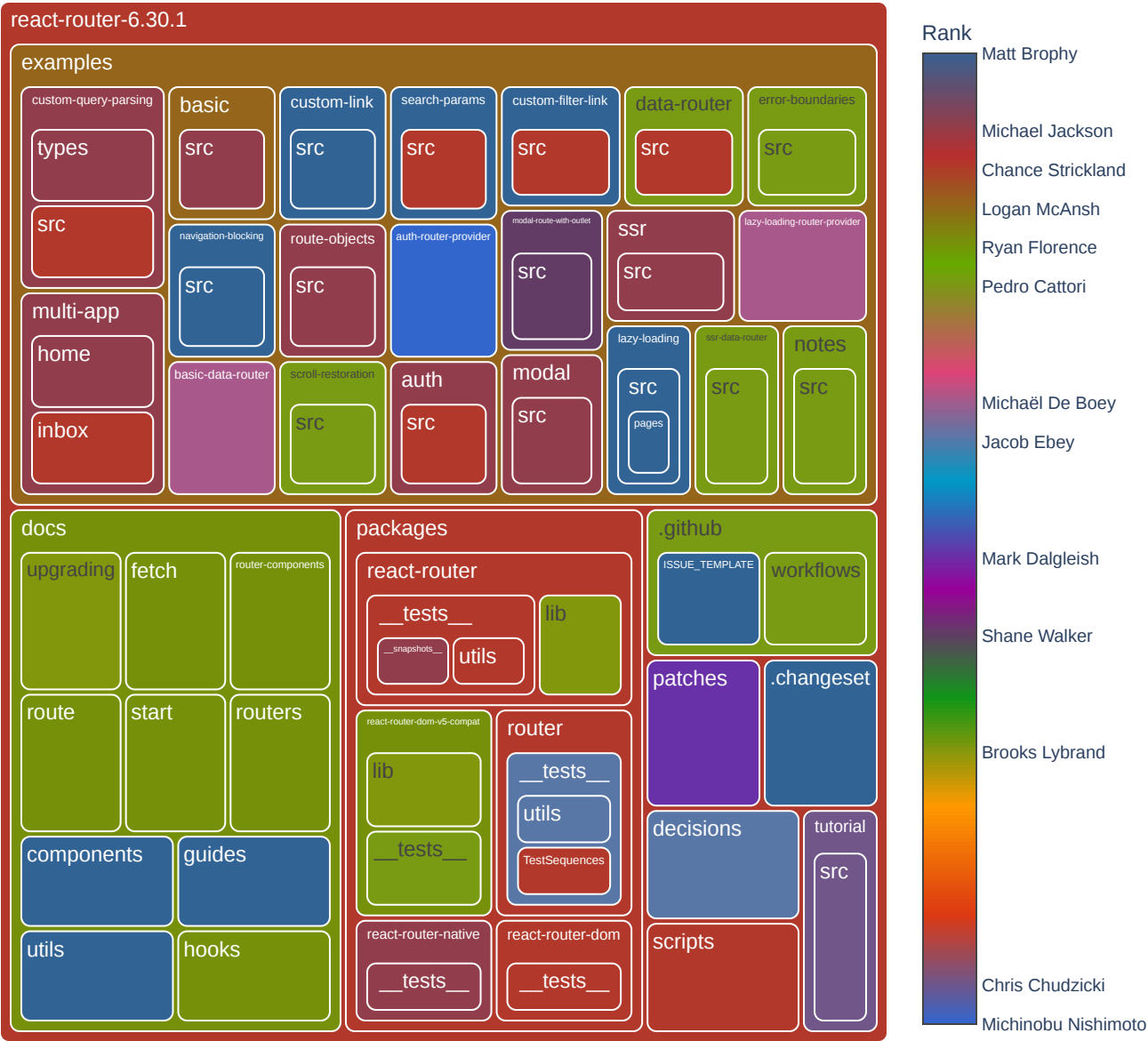
Main author per directory

Main authors with highest number of commits



Second author per directory

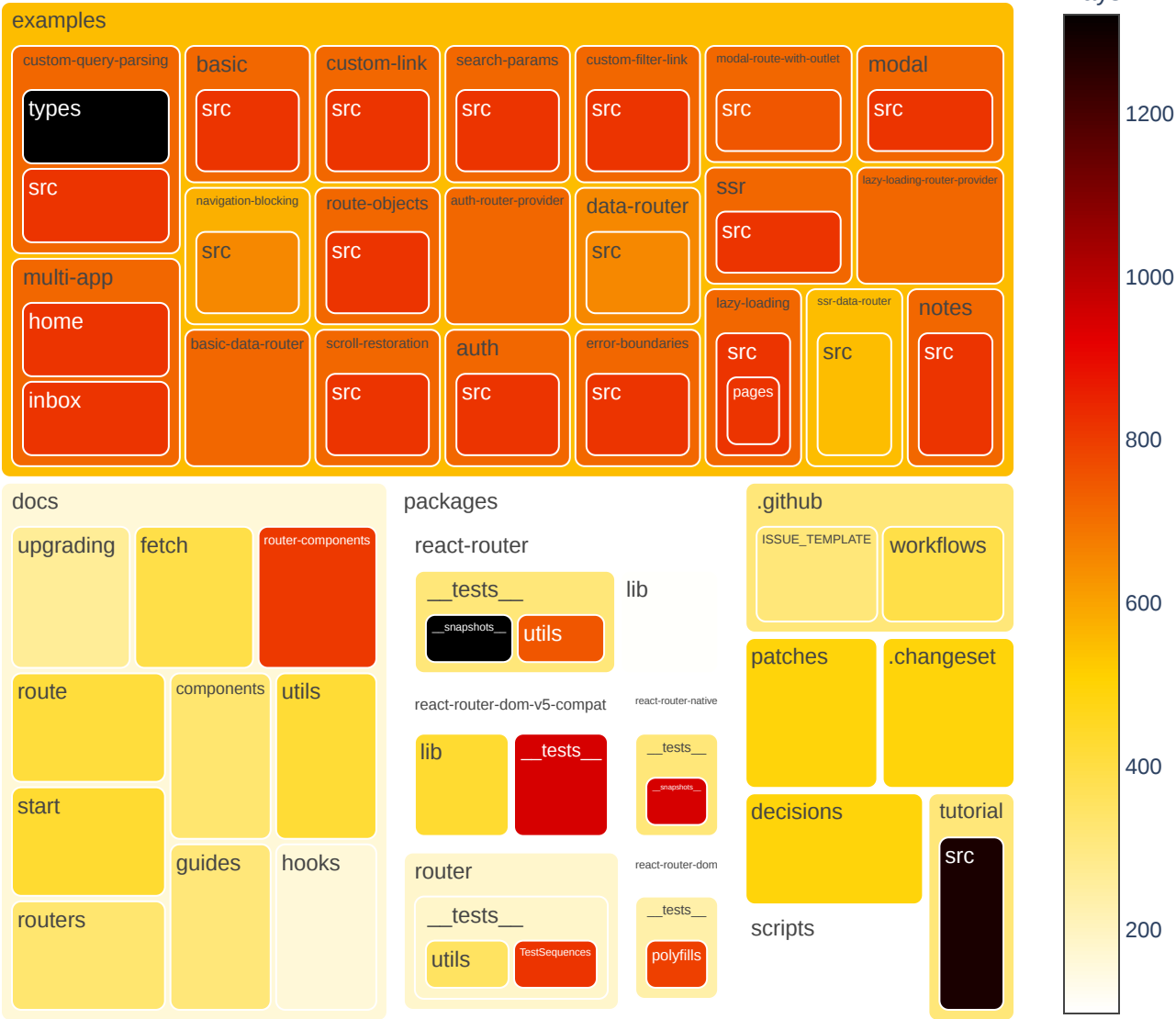
Second author with the second highest number of commits



Days since last commit per directory

Days since last commit

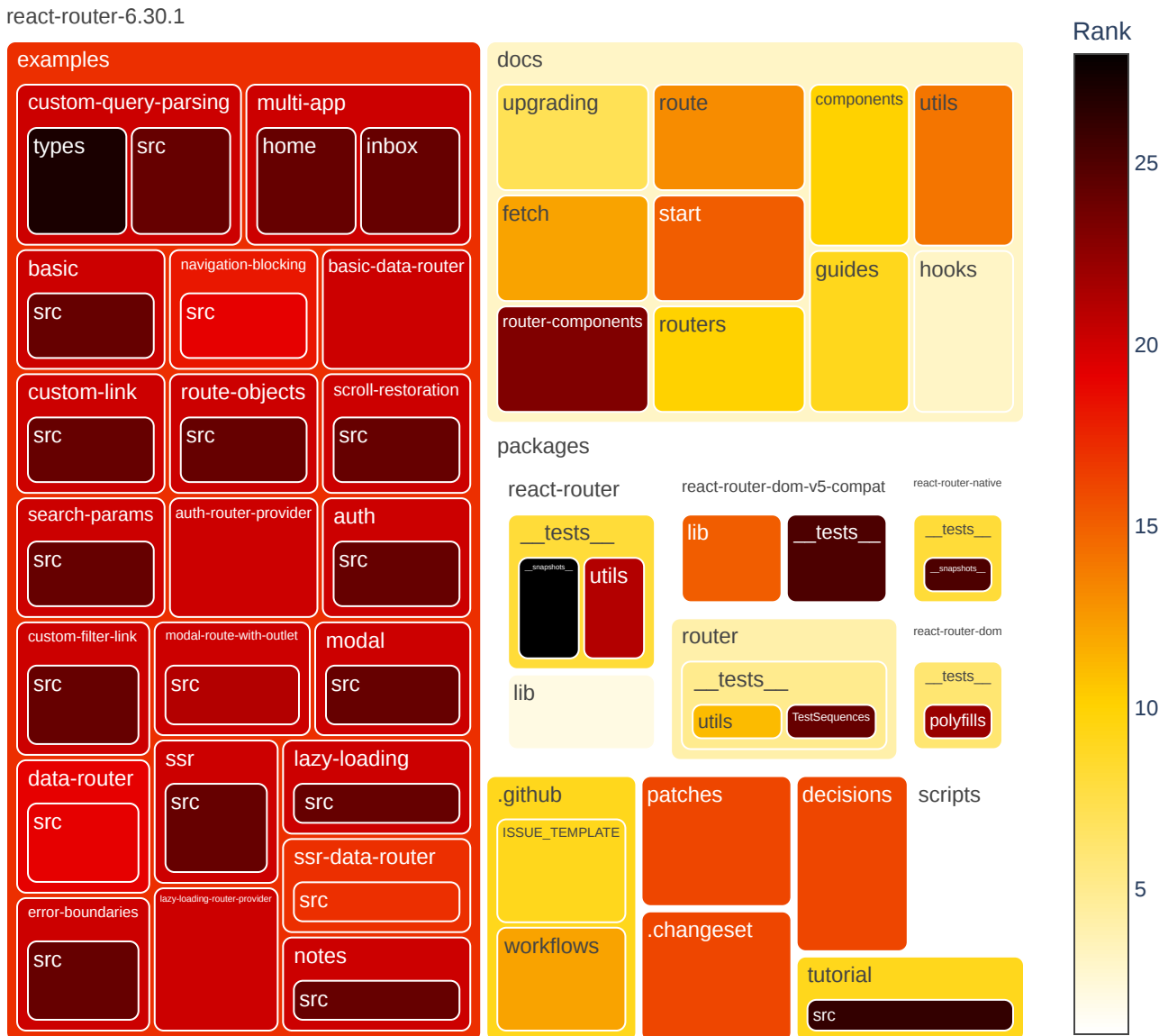
react-router-6.30.1



Days since last commit per directory (ranked)

Rank of days since last commit

react-router-6.30.1



Days since last file creation per directory

Days since last file creation

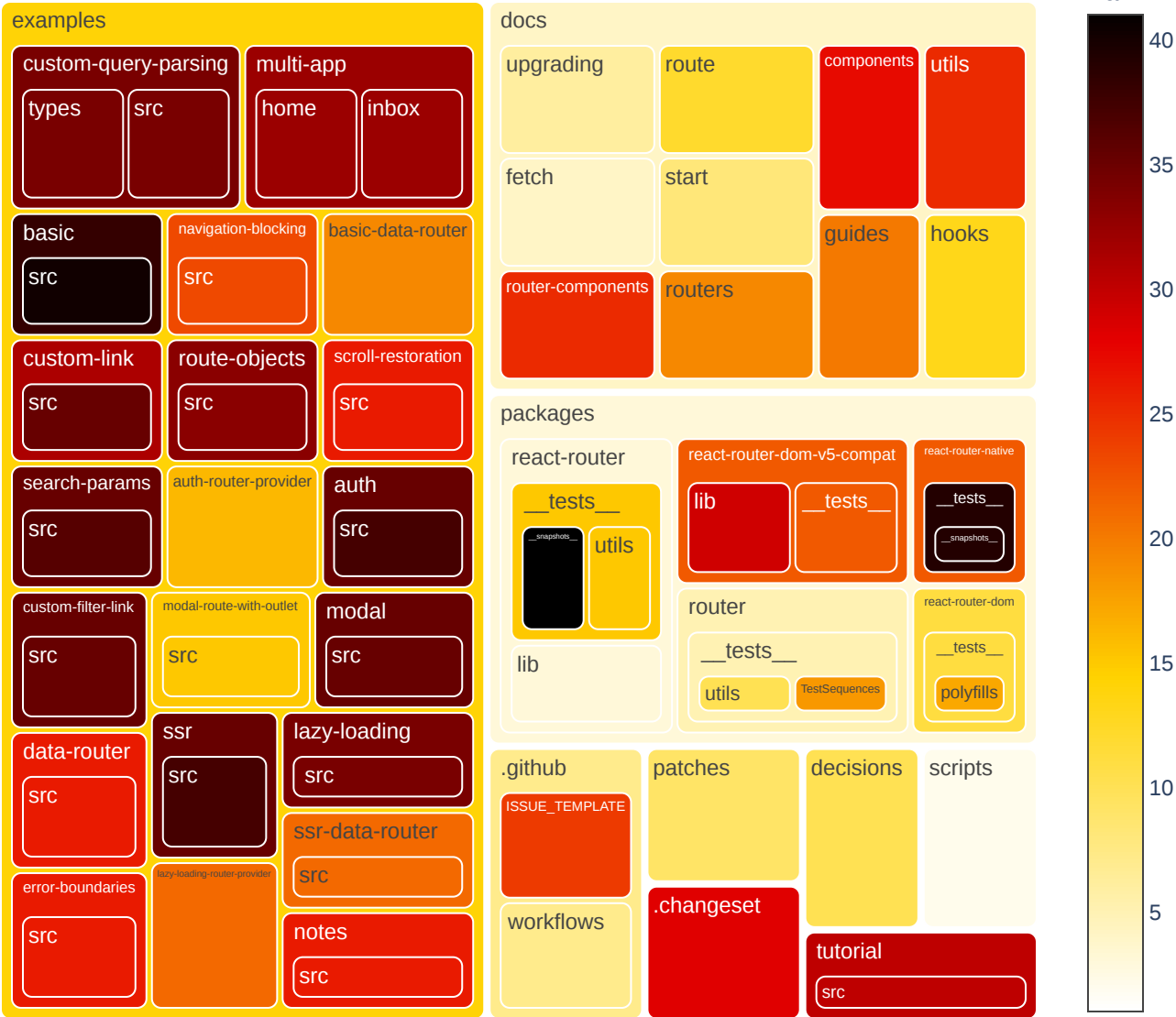
react-router-6.30.1



Days since last file creation per directory (ranked)

Rank of days since last file creation

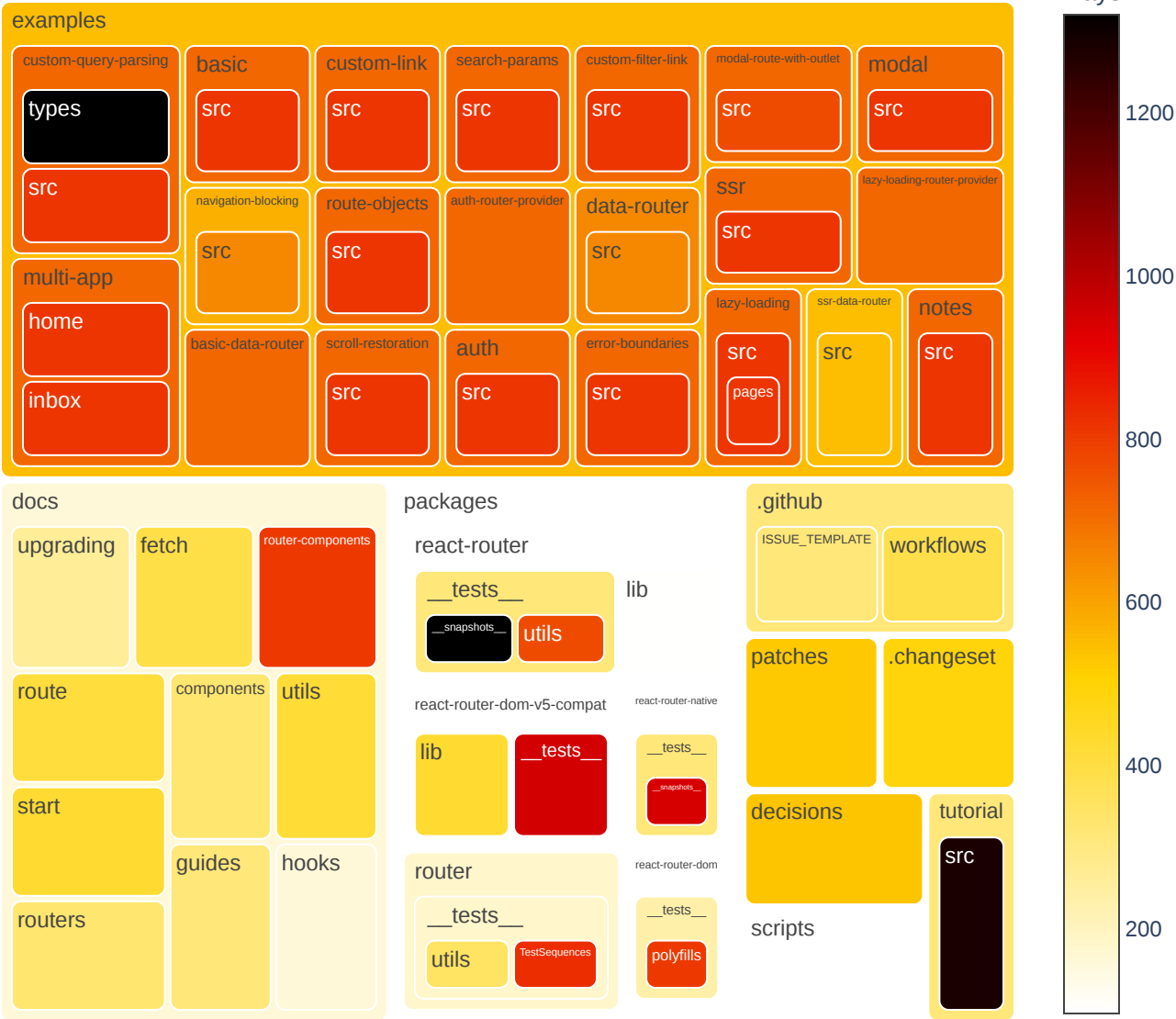
react-router-6.30.1



Days since last file modification per directory

Days since last file modification

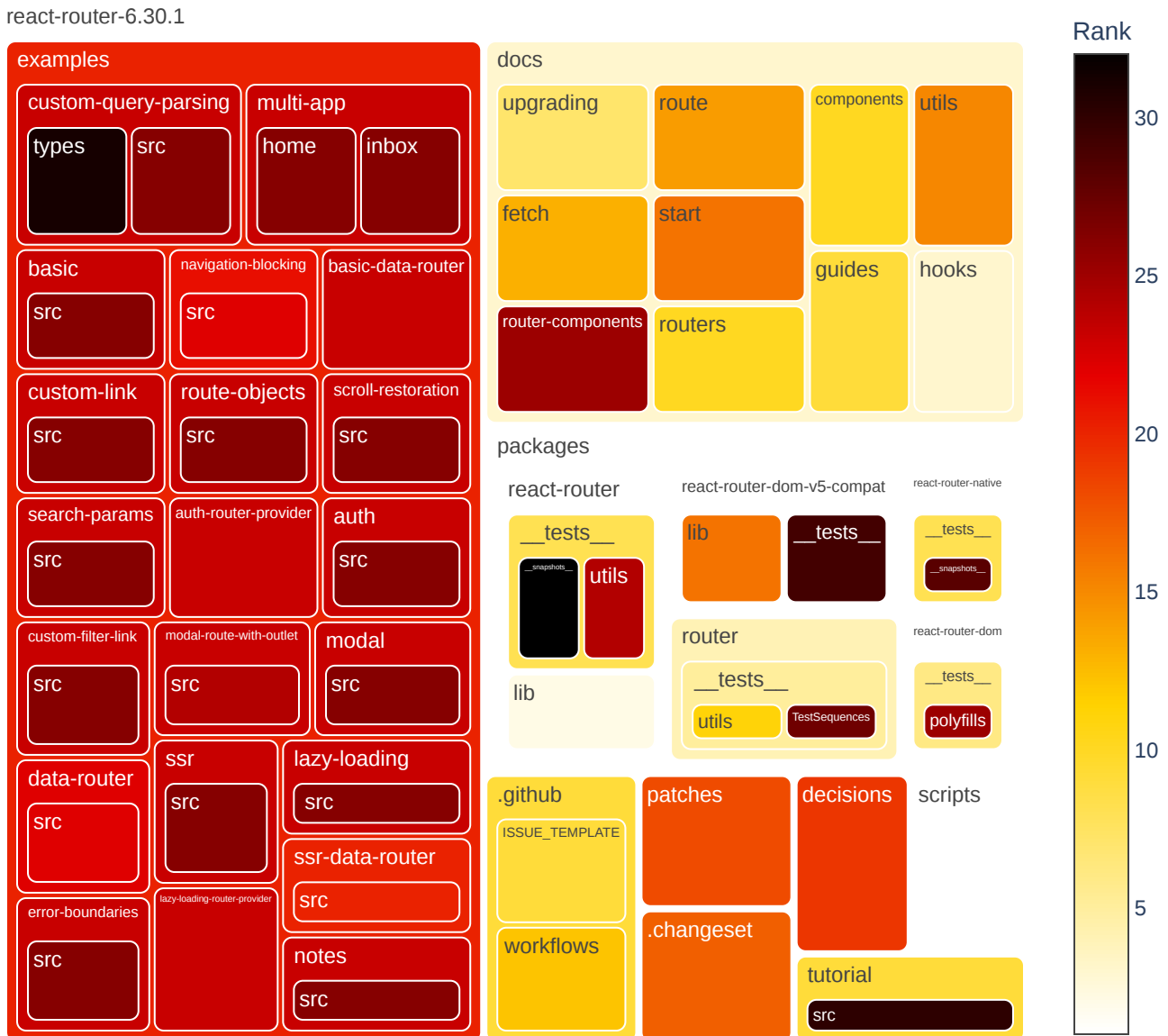
react-router-6.30.1



Days since last file modification per directory (ranked)

Rank of days since last file modification

react-router-6.30.1



Filecount per commit

Shows how many commits had changed one file, how many had changed two files, and so on. The chart is limited to 30 lines for improved readability. The data preview also includes overall statistics including the number of commits that are filtered out in the chart.

Preview data

Sum of commits that changed more than 30 files (each) = 229

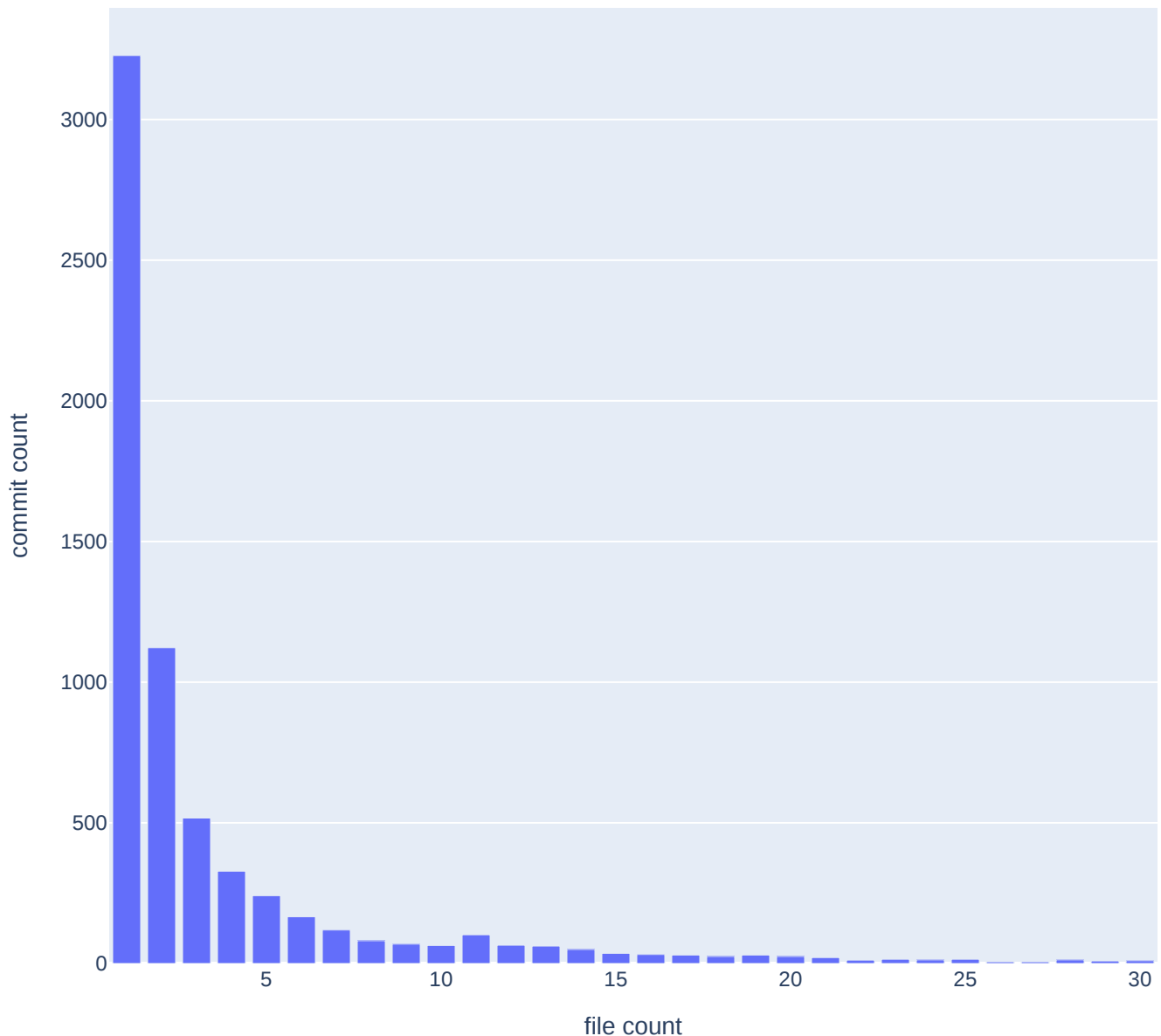
Max changed files with one commit = 498

	filesPerCommit	commitCount
count	117.000000	117.000000
mean	79.179487	57.606838
std	78.464424	318.984216
min	1.000000	1.000000
25%	30.000000	1.000000
50%	59.000000	3.000000
75%	96.000000	8.000000
max	498.000000	3227.000000

	filesPerCommit	commitCount
0	1	3227
1	2	1122
2	3	517
3	4	328
4	5	241
5	6	166
6	7	119
7	8	81
8	9	69
9	10	64
10	11	102
11	12	65
12	13	62
13	14	50
14	15	36
15	16	32
16	17	30
17	18	25
18	19	30
19	20	25
20	21	21
21	22	12
22	23	15
23	24	13
24	25	15
25	26	6
26	27	6
27	28	13
28	29	9
29	30	10

Bar chart with the number of files per commit distribution

Changed files per commit



Pairwise Changed Files

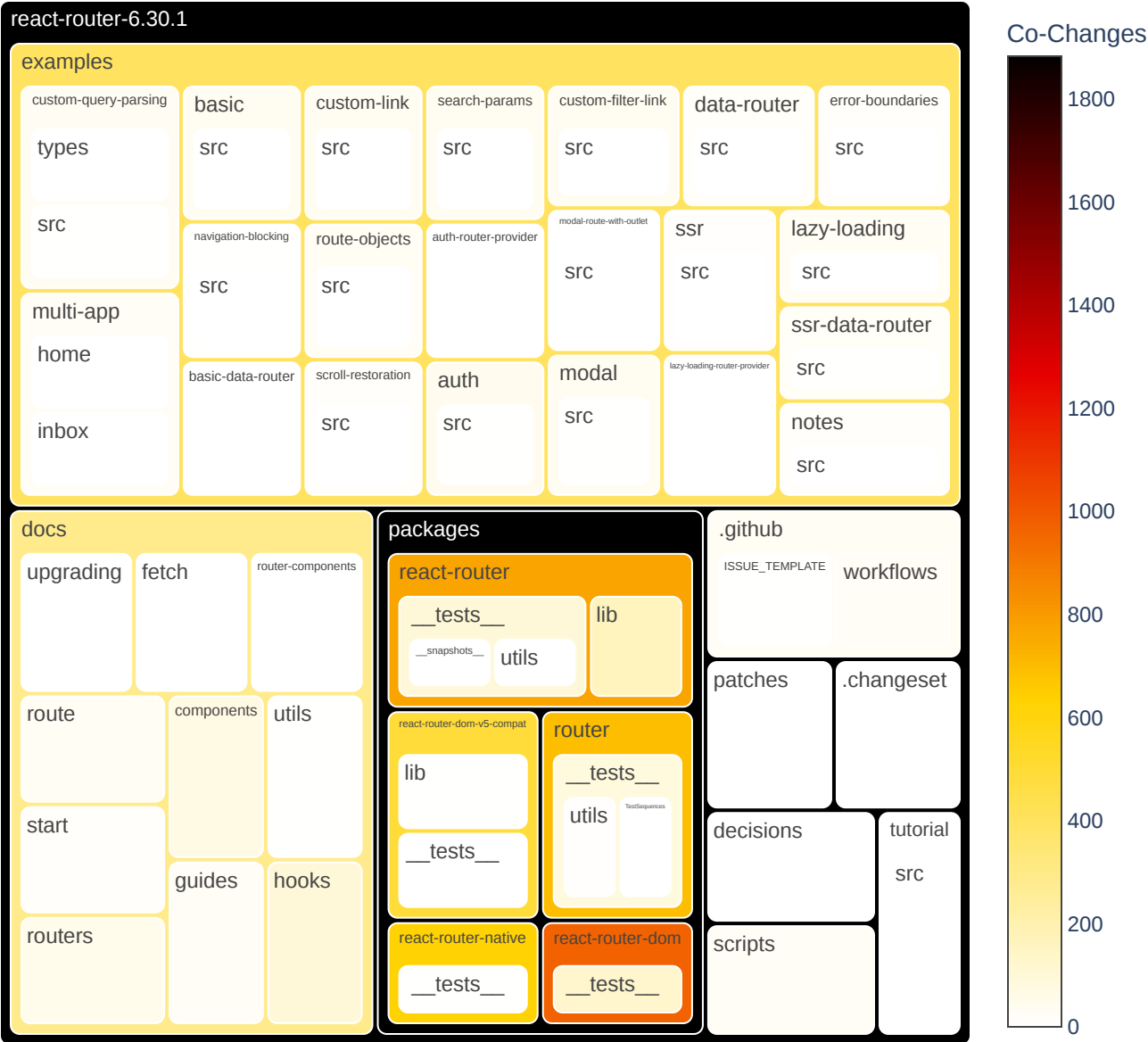
This section analyzes files that were changed together within the same commit and provides several metrics to quantify the strength of the co-change relationship:

- **Commit Count:** The number of commits in which two files were changed together.
- **Commit Lift:** A ratio that indicates whether the co-change pattern is stronger than random chance, given how often each file changes.
- **Jaccard Similarity:** The ratio of commits involving either file that also involved both files.

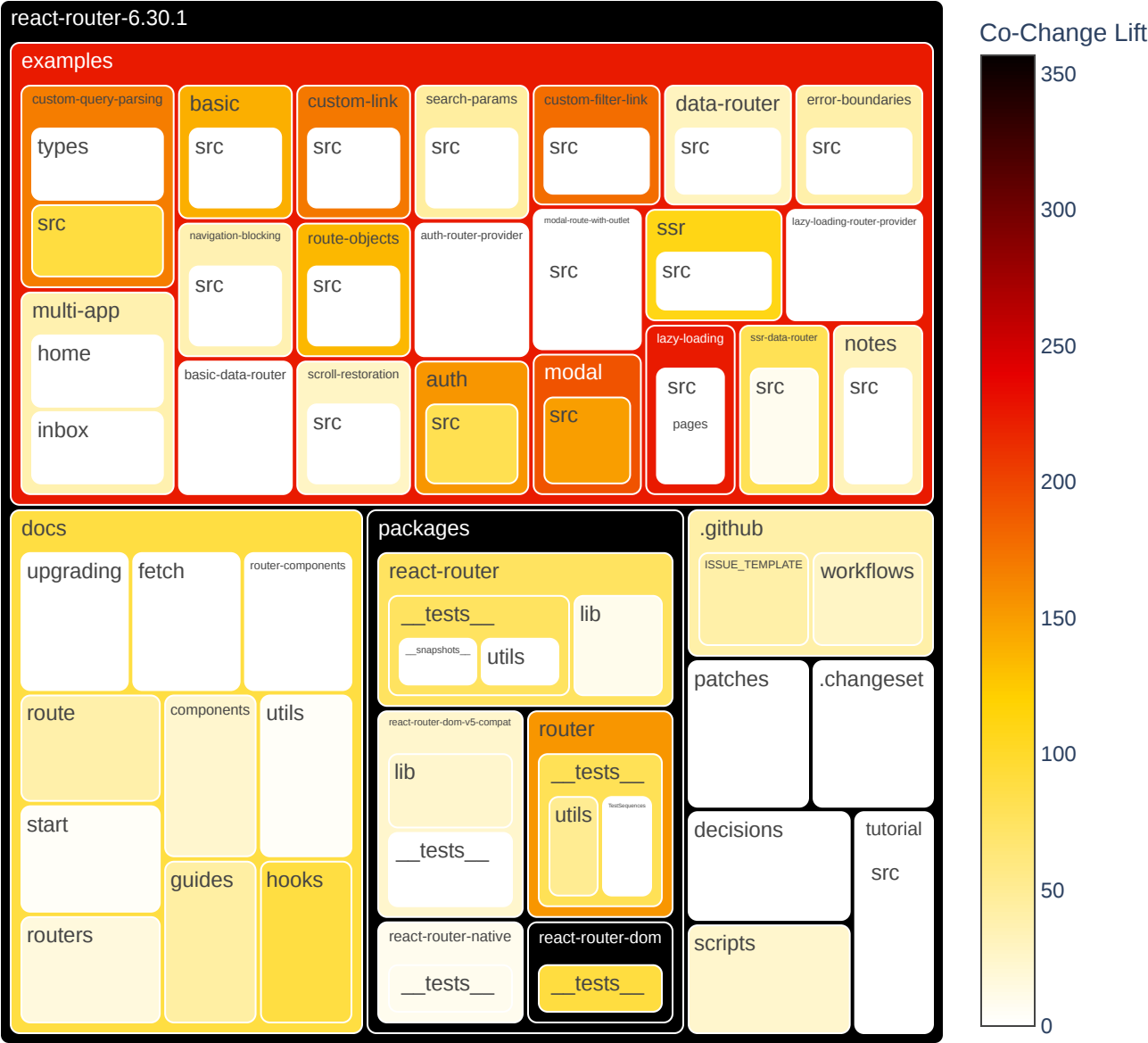
The following tables show the top pairwise changed files based on these metrics. The following charts show how these metrics are distributed across pairs of files that were changed together.

Treemap with files changed frequently with others

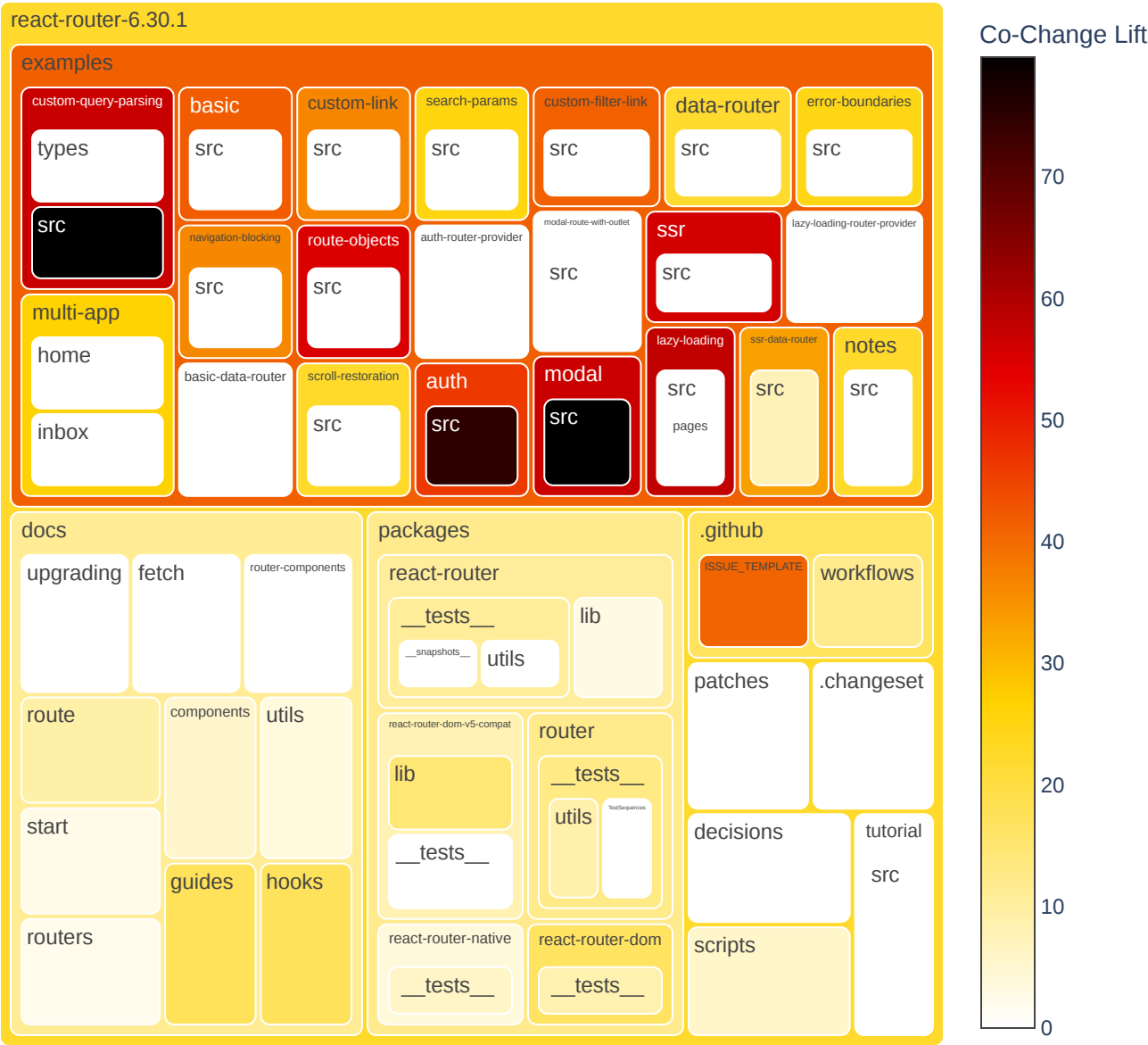
Files that likely co-change with others in update commits



Co-Changing files in update commits max lift (1=random, >1=more than random, <1=



Co-Changing files in update commits average lift (1=random, >1=more than random,

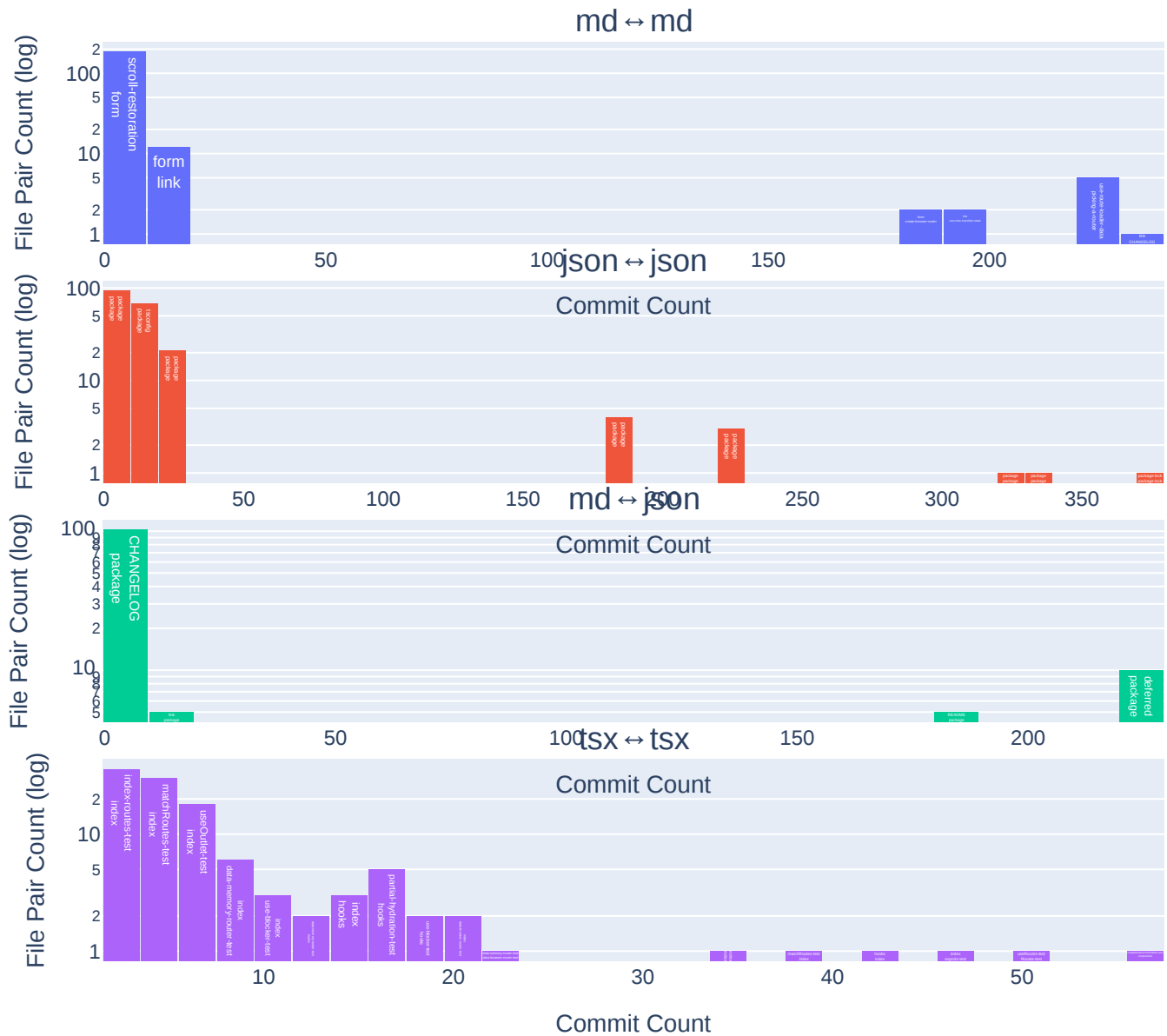


Files changed together by commit count

	fileExtensionPair	updateCommitCount	GroupRank	filePair	filePairWithRelativePath
0	json↔json	376	1	package↔package	packages/react-router-dom/package.json↔packages/react-router/package.json
1	json↔json	332	2	package↔package	packages/react-router-dom/package.json↔packages/react-router-native/package.json
2	json↔json	327	3	package↔package	packages/react-router-native/package.json↔packages/react-router/package.json
3	json↔json	229	4	package↔package	packages/react-router-dom-v5-compat/package.json↔packages/react-router-dom/package.json
4	json↔json	185	5	package↔package	packages/react-router-dom-v5-compat/package.json↔packages/router/package.json
5	json↔json	184	6	package↔package	packages/react-router-native/package.json↔packages/router/package.json
6	json↔json	21	7	package↔package	examples/auth/package.json↔examples/basic/package.json
7	json↔json	20	8	package↔package	examples/auth/package.json↔examples/custom-filter-link/package.json
8	json↔json	19	9	package↔package	packages/react-router/package.json↔examples/auth/package.json
9	json↔json	18	10	package↔package	packages/react-router/package.json↔examples/custom-filter-link/package.json
10	md↔json	224	1	CHANGELOG↔package	packages/react-router-dom/CHANGELOG.md↔packages/react-router-dom/package.json
11	md↔json	223	2	CHANGELOG↔package	packages/react-router-dom-v5-compat/CHANGELOG.md↔packages/react-router-dom-v5-compat/package.json
12	md↔json	181	3	CHANGELOG↔package	packages/react-router-dom/CHANGELOG.md↔packages/router/package.json
13	md↔json	180	4	CHANGELOG↔package	packages/react-router-native/CHANGELOG.md↔packages/router/package.json
14	md↔json	10	5	create-browser-router↔package	docs/routers/create-browser-router.md↔packages/react-router-dom-v5-compat/package.json
15	md↔json	5	6	link↔package	docs/components/link.md↔packages/react-router-dom-v5-compat/package.json
16	md↔json	4	7	picking-a-router↔package	docs/routers/picking-a-router.md↔packages/react-router-dom-v5-compat/package.json
17	md↔json	3	8	use-submit↔package	docs/hooks/use-submit.md↔packages/react-router-dom-v5-compat/package.json
18	md↔md	235	1	CHANGELOG↔CHANGELOG	packages/react-router-dom/CHANGELOG.md↔packages/react-router/CHANGELOG.md
19	md↔md	229	2	CHANGELOG↔CHANGELOG	packages/react-router-dom-v5-compat/CHANGELOG.md↔packages/react-router-dom/CHANGELOG.md
20	md↔md	195	3	CHANGELOG↔CHANGELOG	packages/react-router/CHANGELOG.md↔packages/router/CHANGELOG.md
21	md↔md	193	4	CHANGELOG↔CHANGELOG	packages/react-router-dom/CHANGELOG.md↔packages/router/CHANGELOG.md
22	md↔md	187	5	CHANGELOG↔CHANGELOG	packages/react-router-native/CHANGELOG.md↔packages/router/CHANGELOG.md
23	md↔md	12	6	use-fetcher↔use-submit	docs/hooks/use-fetcher.md↔docs/hooks/use-submit.md
24	md↔md	11	7	link↔use-navigate	docs/components/link.md↔docs/hooks/use-navigate.md
25	md↔md	10	8	CHANGELOG↔DEVELOPMENT	packages/react-router-native/CHANGELOG.md↔DEVELOPMENT.md
26	md↔md	9	9	form↔use-navigate	docs/components/form.md↔docs/hooks/use-navigate.md
27	md↔md	8	10	form↔link	docs/components/form.md↔docs/components/link.md
28	tsx↔tsx	57	1	index↔index	packages/react-router-native/index.tsx↔packages/react-router-dom/index.tsx
29	tsx↔tsx	50	2	data-browser-router-test↔index	packages/react-router-dom/__tests__/data-browser-router-test.tsx↔packages/react-router-dom/index.tsx
30	tsx↔tsx	47	3	index↔components	packages/react-router-dom/index.tsx↔packages/react-router/lib/components.tsx
31	tsx↔tsx	42	4	index↔server	packages/react-router-dom/index.tsx↔packages/react-router-dom/server.tsx
32	tsx↔tsx	39	5	hooks↔index	packages/react-router/lib/hooks.tsx↔packages/react-router-dom/index.tsx
33	tsx↔tsx	34	6	hooks↔components	packages/react-router/lib/hooks.tsx↔packages/react-router/lib/components.tsx
34	tsx↔tsx	22	7	components↔server	packages/react-router/lib/components.tsx↔packages/react-router-dom/server.tsx
35	tsx↔tsx	21	8	hooks↔server	packages/react-router/lib/hooks.tsx↔packages/react-router-dom/server.tsx

	fileExtensionPair	updateCommitCount	GroupRank	filePair	filePairWithRelativePath
36	tsx↔tsx	20	9	index↔nav-link-active-test	packages/react-router-dom/index.tsx↔packages/react-router-dom/_tests_/nav-link-active-test.tsx
37	tsx↔tsx	19	10	hooks↔data-browser-router-test	packages/react-router/lib/hooks.tsx↔packages/react-router-dom/_tests_/data-browser-router-test.tsx

Co-Changed Files by their commit count



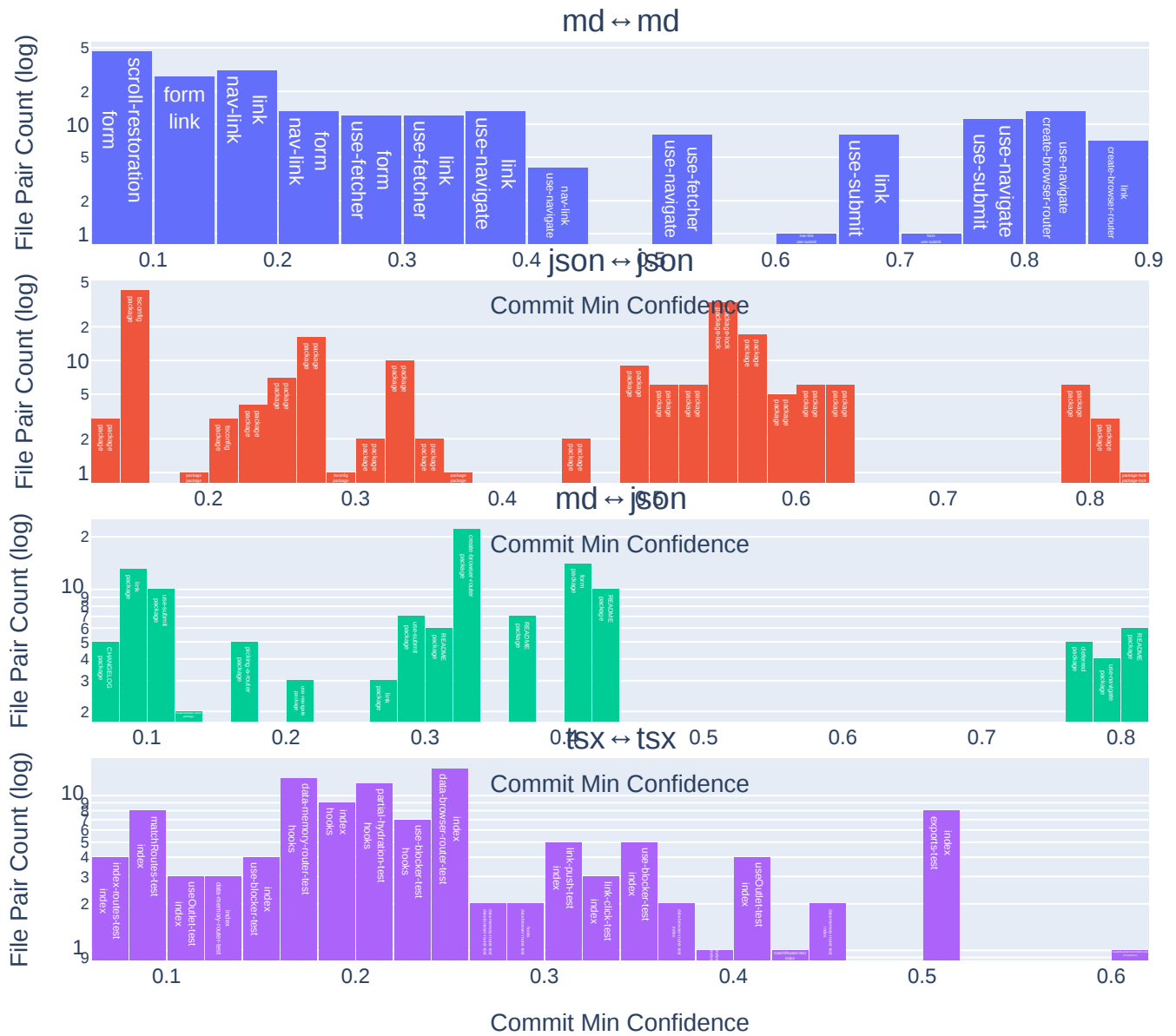
Files changed together by commit min confidence

The commit min confidence is the commit count where both files were changed divided by the commit count of the file with the least commits. This metric is useful to identify pairs of files that are frequently changed together and is not biased by single files that are changed very often.

	fileExtensionPair	updateCommitMinConfidence	GroupRank	filePair	filePairWithRelativePath
0	json↔json	0.833703	1	package↔package	packages/react-router-dom/package.json↔packages/react-router/package.json
1	json↔json	0.809187	2	package↔package	packages/react-router-dom-v5-compat/package.json↔packages/react-router-dom/package.json
2	json↔json	0.794258	3	package↔package	packages/react-router-dom/package.json↔packages/react-router-native/package.json
3	json↔json	0.793991	4	package↔package	packages/react-router-dom-v5-compat/package.json↔packages/router/package.json
4	json↔json	0.789700	5	package↔package	packages/react-router-native/package.json↔packages/router/package.json
5	json↔json	0.782297	6	package↔package	packages/react-router-native/package.json↔packages/react-router/package.json
6	json↔json	0.625000	7	package↔package	examples/auth/package.json↔examples/custom-filter-link/package.json
7	json↔json	0.600000	8	package↔package	examples/auth/package.json↔examples/basic/package.json
8	json↔json	0.588235	9	package↔package	examples/basic/package.json↔examples/custom-link/package.json
9	json↔json	0.571429	10	package↔package	examples/auth/package.json↔examples/modal/package.json
10	md↔json	0.813869	1	CHANGELOG↔package	packages/react-router-dom-v5-compat/CHANGELOG.md↔packages/react-router-dom-v5-compat/package.json
11	md↔json	0.794326	2	CHANGELOG↔package	packages/react-router-dom/CHANGELOG.md↔packages/react-router-dom/package.json
12	md↔json	0.780488	3	CHANGELOG↔package	packages/react-router/CHANGELOG.md↔packages/react-router/package.json
13	md↔json	0.776824	4	CHANGELOG↔package	packages/react-router-dom/CHANGELOG.md↔packages/router/package.json
14	md↔json	0.772532	5	CHANGELOG↔package	packages/react-router-native/CHANGELOG.md↔packages/router/package.json
15	md↔json	0.428571	6	README↔package	examples/modal/README.md↔packages/react-router-dom/package.json
16	md↔json	0.400000	7	README↔package	examples/search-params/README.md↔packages/react-router-dom/package.json
17	md↔json	0.363636	8	README↔package	examples/basic/README.md↔packages/react-router-dom/package.json
18	md↔json	0.333333	9	use-route-loader-data↔package	docs/hooks/use-route-loader-data.md↔packages/react-router-dom-v5-compat/package.json
19	md↔json	0.300000	10	README↔package	examples/auth/README.md↔examples/custom-filter-link/package.json
20	md↔md	0.888889	1	README↔README	examples/custom-filter-link/README.md↔examples/custom-link/README.md
21	md↔md	0.857143	2	README↔README	examples/custom-filter-link/README.md↔examples/modal/README.md
22	md↔md	0.835766	3	CHANGELOG↔CHANGELOG	packages/react-router-dom-v5-compat/CHANGELOG.md↔packages/react-router-dom/CHANGELOG.md
23	md↔md	0.833333	4	CHANGELOG↔CHANGELOG	packages/react-router-dom/CHANGELOG.md↔packages/react-router/CHANGELOG.md
24	md↔md	0.812500	5	CHANGELOG↔CHANGELOG	packages/react-router-router/CHANGELOG.md↔packages/router/CHANGELOG.md
25	md↔md	0.804167	6	CHANGELOG↔CHANGELOG	packages/react-router-dom/CHANGELOG.md↔packages/router/CHANGELOG.md
26	md↔md	0.800000	7	README↔README	examples/ssr/README.md↔examples/auth/README.md
27	md↔md	0.779167	8	CHANGELOG↔CHANGELOG	packages/react-router-native/CHANGELOG.md↔packages/router/CHANGELOG.md
28	md↔md	0.750000	9	README↔README	examples/custom-filter-link/README.md↔examples/route-objects/README.md
29	md↔md	0.727273	10	README↔README	examples/ssr/README.md↔examples/basic/README.md
30	tsx↔tsx	0.600000	1	index↔index	packages/react-router-native/index.tsx↔packages/react-router-dom/index.tsx
31	tsx↔tsx	0.500000	2	partial-hydration-test↔hooks	packages/react-router-dom/_tests_/partial-hydration-test.tsx↔packages/react-router/lib/hooks.tsx
32	tsx↔tsx	0.454545	3	data-browser-router-test↔index	packages/react-router-dom/_tests_/data-browser-router-test.tsx↔packages/react-router-dom/index.tsx
33	tsx↔tsx	0.437500	4	exports-test↔exports-test	packages/react-router-dom/_tests_/exports-test.tsx↔packages/react-router-native/_tests_/exports-test.tsx
34	tsx↔tsx	0.419643	5	index↔components	packages/react-router-dom/index.tsx↔packages/react-router/lib/components.tsx

	fileExtensionPair	updateCommitMinConfidence	GroupRank	filePair	filePairWithRelativePath
35	tsx↔tsx	0.407767	6	index↔server	packages/react-router-dom/index.tsx↔packages/react-router-dom/server.tsx
36	tsx↔tsx	0.400000	7	server↔data-static-router-test	packages/react-router-dom/server.tsx↔packages/react-router-dom/_tests_/data-static-router-test.tsx
37	tsx↔tsx	0.388889	8	index↔exports-test	packages/react-router-dom/index.tsx↔packages/react-router-dom/_tests_/exports-test.tsx
38	tsx↔tsx	0.375000	9	index↔exports-test	packages/react-router-dom/index.tsx↔packages/react-router-native/_tests_/exports-test.tsx
39	tsx↔tsx	0.363636	10	link-push-test↔index	packages/react-router-dom/_tests_/link-push-test.tsx↔packages/react-router-dom/index.tsx

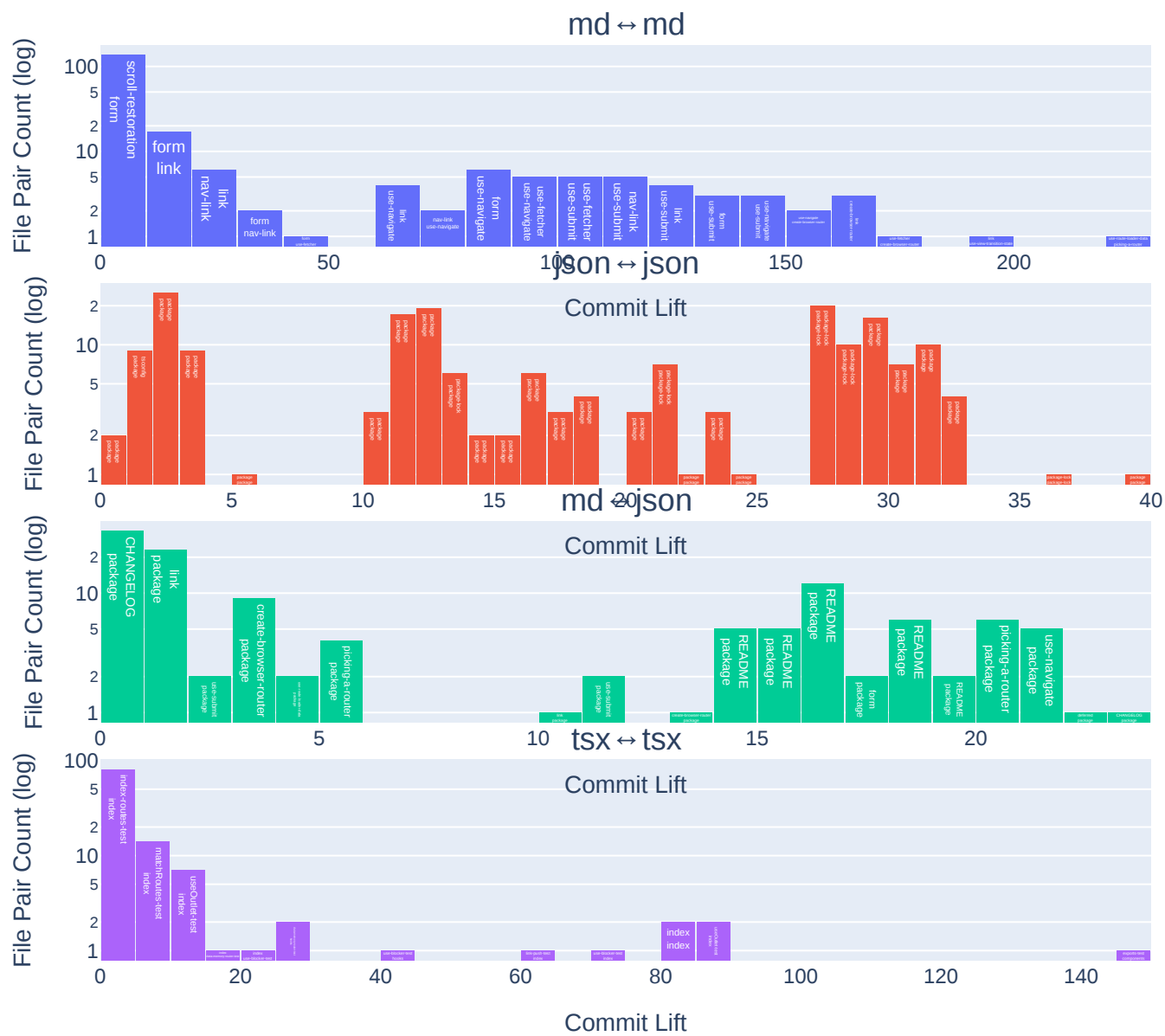
Co-Changed Files by their commit min confidence



Files changed together by commit lift

	fileExtensionPair	updateCommitLift	GroupRank	filePair	filePairWithRelativePath
0	json↔json	39.622222	1	package↔package	examples/error-boundaries/package.json↔examples/notes/package.json
1	json↔json	36.020202	2	package-lock↔package	examples/navigation-blocking/package-lock.json↔examples/navigation-blocking/package.json
2	json↔json	32.775735	3	package↔package	examples/custom-filter-link/package.json↔examples/custom-link/package.json
3	json↔json	32.592473	4	package↔package	examples/lazy-loading/package.json↔examples/multi-app/package.json
4	json↔json	31.839286	5	package↔package	examples/auth/package.json↔examples/custom-filter-link/package.json
5	json↔json	31.697778	6	package↔package	examples/notes/package.json↔examples/scroll-restoration/package.json
6	json↔json	31.573958	7	package↔package	examples/custom-filter-link/package.json↔examples/multi-app/package.json
7	json↔json	31.541103	8	package↔package	examples/custom-query-parsing/package.json↔examples/lazy-loading/package.json
8	json↔json	30.954861	9	package↔package	examples/basic/package.json↔examples/custom-filter-link/package.json
9	json↔json	30.565714	10	package↔package	examples/auth/package.json↔examples/search-params/package.json
10	md↔json	23.879464	1	README↔package	examples/modal/README.md↔examples/custom-filter-link/package.json
11	md↔json	22.474790	2	README↔package	examples/modal/README.md↔examples/custom-link/package.json
12	md↔json	21.832653	3	README↔package	examples/modal/README.md↔examples/auth/package.json
13	md↔json	21.226190	4	README↔package	examples/modal/README.md↔examples/basic/package.json
14	md↔json	20.377143	5	README↔package	examples/auth/README.md↔examples/auth/package.json
15	md↔json	19.811111	6	README↔package	examples/auth/README.md↔examples/basic/package.json
16	md↔json	18.572917	7	README↔package	examples/custom-filter-link/README.md↔examples/custom-filter-link/package.json
17	md↔json	18.524675	8	README↔package	examples/basic/README.md↔examples/auth/package.json
18	md↔json	18.010101	9	README↔package	examples/basic/README.md↔examples/basic/package.json
19	md↔json	17.480392	10	README↔package	examples/custom-filter-link/README.md↔examples/custom-link/package.json
20	md↔md	222.875000	1	README↔README	examples/lazy-loading/README.md↔examples/route-objects/README.md
21	md↔md	191.035714	2	README↔README	examples/modal/README.md↔examples/route-objects/README.md
22	md↔md	176.098765	3	README↔README	examples/custom-filter-link/README.md↔examples/custom-link/README.md
23	md↔md	169.809524	4	README↔README	examples/custom-filter-link/README.md↔examples/modal/README.md
24	md↔md	167.156250	5	README↔README	examples/custom-query-parsing/README.md↔examples/route-objects/README.md
25	md↔md	152.828571	6	README↔README	examples/auth/README.md↔examples/modal/README.md
26	md↔md	148.583333	7	README↔README	examples/custom-filter-link/README.md↔examples/route-objects/README.md
27	md↔md	142.640000	8	README↔README	examples/auth/README.md↔examples/search-params/README.md
28	md↔md	138.935065	9	README↔README	examples/basic/README.md↔examples/modal/README.md
29	md↔md	133.725000	10	README↔README	examples/auth/README.md↔examples/route-objects/README.md
30	tsx↔tsx	148.583333	1	App↔App	examples/modal/src/App.tsx↔examples/search-params/src/App.tsx
31	tsx↔tsx	89.150000	2	App↔App	examples/custom-query-parsing/src/App.tsx↔examples/modal/src/App.tsx
32	tsx↔tsx	81.045455	3	App↔App	examples/auth/src/App.tsx↔examples/modal/src/App.tsx
33	tsx↔tsx	74.291667	4	descendant-routes-splat-matching-test↔layout-routes-test	packages/react-router/__tests__/descendant-routes-splat-matching-test.tsx↔packages/react-router/__tests__/layout-routes-test.tsx
34	tsx↔tsx	64.836364	5	App↔App	examples/auth/src/App.tsx↔examples/custom-query-parsing/src/App.tsx
35	tsx↔tsx	43.336806	6	exports-test↔exports-test	packages/react-router-dom/__tests__/exports-test.tsx↔packages/react-router-native/__tests__/exports-test.tsx
36	tsx↔tsx	27.859375	7	index-routes-test↔matchRoutes-test	packages/react-router/__tests__/index-routes-test.tsx↔packages/react-router/__tests__/matchRoutes-test.tsx
37	tsx↔tsx	25.716346	8	matchRoutes-test↔useMatch-test	packages/react-router/__tests__/matchRoutes-test.tsx↔packages/react-router/__tests__/useMatch-test.tsx
38	tsx↔tsx	23.256522	9	components↔static-location-test	packages/react-router-dom-v5-compat/lib/components.tsx↔packages/react-router-dom/__tests__/static-location-test.tsx
39	tsx↔tsx	19.380435	10	useRoutes-test↔Routes-test	packages/react-router/__tests__/useRoutes-test.tsx↔packages/react-router/__tests__/Routes-test.tsx

Co-Changed Files by their commit lift

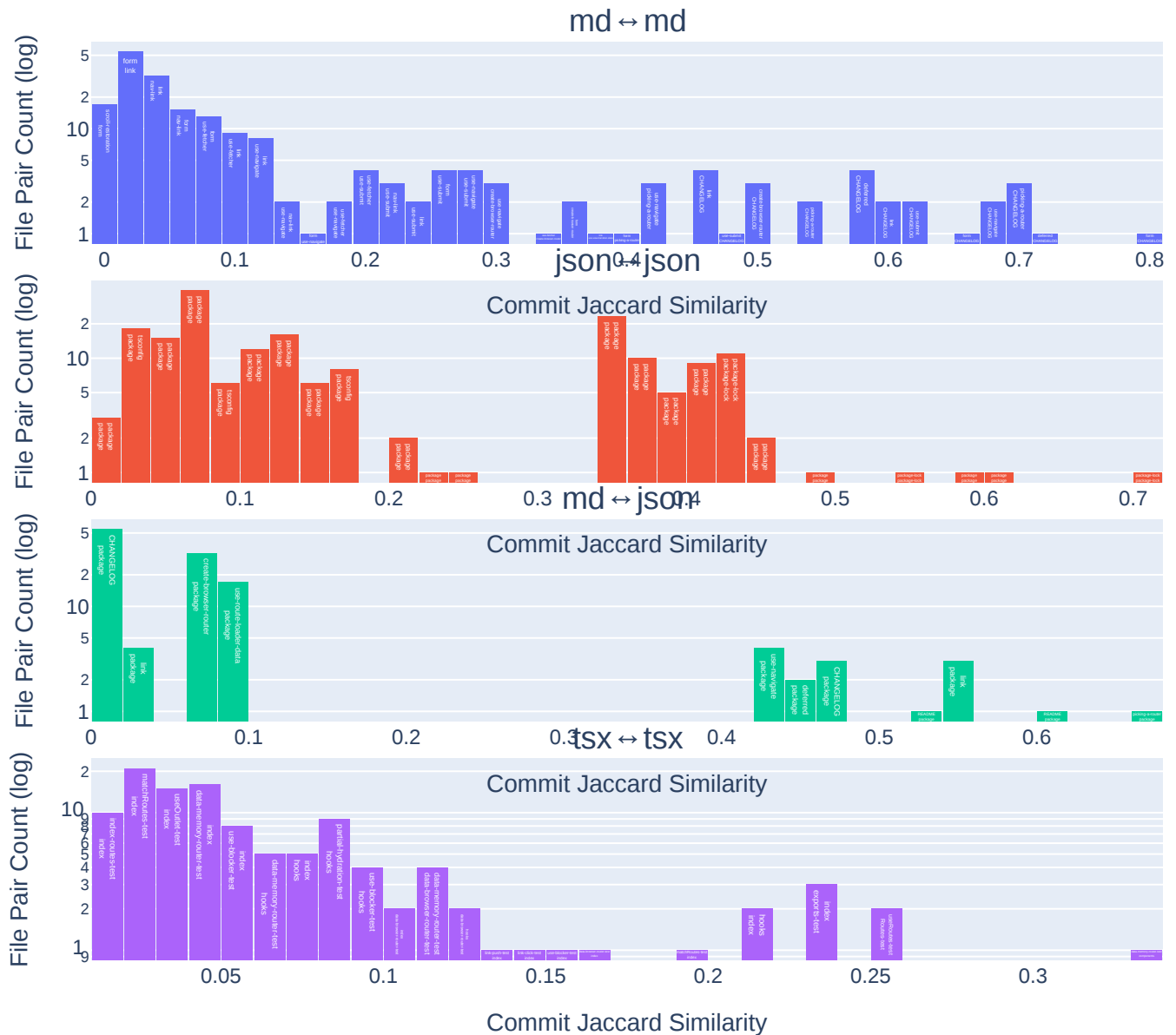


Files changed together by commit Jaccard similarity

	fileExtensionPair	updateCommitJaccardSimilarity	GroupRank	filePair	filePairWithRelative
0	json↔json	0.708098	1	package↔package	packages/react-router-dom/package.json↔packages/r outer/package
1	json↔json	0.618250	2	package↔package	packages/react-router-dom/package.json↔packages/r outer-native/package
2	json↔json	0.597806	3	package↔package	packages/react-router-native/package.json↔packages/r outer/package
3	json↔json	0.558912	4	package↔package	packages/react-router-dor compat/package.json↔packages/router/package
4	json↔json	0.485169	5	package↔package	packages/react-router-dor compat/package.json↔packages/react-rc native/package
5	json↔json	0.453465	6	package↔package	packages/react-router-dor compat/package.json↔packages/react-router-dom/package
6	json↔json	0.449020	7	package↔package	packages/react-router-dor compat/package.json↔packages/react-router/package
7	json↔json	0.434783	8	package↔package	examples/custom-filter-link/package.json↔examples/cus link/package
8	json↔json	0.428571	9	package↔package	examples/auth/package.json↔examples/se params/package
9	json↔json	0.425532	10	package↔package	examples/auth/package.json↔examples/custom-l ink/package
10	md↔json	0.667665	1	CHANGELOG↔package	packages/react-router-dor compat/CHANGELOG.md↔packages/react-router-dor compat/package
11	md↔json	0.614334	2	CHANGELOG↔package	packages/router/CHANGELOG.md↔packages/router/package
12	md↔json	0.550459	3	CHANGELOG↔package	packages/react-rc native/CHANGELOG.md↔packages/router/package
13	md↔json	0.541916	4	CHANGELOG↔package	packages/react-rc dom/CHANGELOG.md↔packages/router/package
14	md↔json	0.533923	5	CHANGELOG↔package	packages/r outer/CHANGELOG.md↔packages/router/package
15	md↔json	0.475480	6	CHANGELOG↔package	packages/react-router-dor compat/CHANGELOG.md↔packages/react-rc native/package
16	md↔json	0.470588	7	CHANGELOG↔package	packages/react-router-dom/CHANGELOG.md↔packages/r outer-native/package
17	md↔json	0.444223	8	CHANGELOG↔package	packages/react-router-dor compat/CHANGELOG.md↔packages/react-rc dom/package
18	md↔json	0.440079	9	CHANGELOG↔package	packages/react-router-dom/CHANGELOG.md↔packages/r outer-dom/package
19	md↔json	0.439842	10	CHANGELOG↔package	packages/react-router-dor compat/CHANGELOG.md↔packages/react-router/package
20	md↔md	0.800000	1	README↔README	examples/custom-filter-link/README.md↔examples/cus link/README
21	md↔md	0.717868	2	CHANGELOG↔CHANGELOG	packages/react-router-dor compat/CHANGELOG.md↔packages/react-rc native/CHANGELOG
22	md↔md	0.703593	3	CHANGELOG↔CHANGELOG	packages/react-router-dom/CHANGELOG.md↔packages/r outer/CHANGELOG
23	md↔md	0.700306	4	CHANGELOG↔CHANGELOG	packages/react-router-dor compat/CHANGELOG.md↔packages/react-rc dom/CHANGELOG
24	md↔md	0.689759	5	CHANGELOG↔CHANGELOG	packages/react-router-native/CHANGELOG.md↔packages/r outer/CHANGELOG
25	md↔md	0.666667	6	README↔README	examples/auth/README.md↔examples/se params/README
26	md↔md	0.615385	7	README↔README	examples/auth/README.md↔examples/basic/README
27	md↔md	0.600000	8	README↔README	examples/custom-l ink/README.md↔examples/modal/README
28	md↔md	0.587349	9	CHANGELOG↔CHANGELOG	packages/r outer/CHANGELOG.md↔packages/router/CHANGELOG
29	md↔md	0.586626	10	CHANGELOG↔CHANGELOG	packages/react-rc dom/CHANGELOG.md↔packages/router/CHANGELOG
30	tsx↔tsx	0.333333	1	App↔App	examples/modal/src/App.tsx↔examples/se params/src/App
31	tsx↔tsx	0.259259	2	exports-test↔exports-test	packages/react-router-dom/_tests_/exp test.tsx↔packages/react-router-native/_tests_/exports-te
32	tsx↔tsx	0.250000	3	descendant-routes-splat- matching-test↔layout- routes-test	packages/react-router/_tests_/descendant-routes-s matching-test.tsx↔packages/react-router/_tests_/la routes-test
33	tsx↔tsx	0.235294	4	App↔App	examples/auth/src/App.tsx↔examples/custom-q parsing/src/App

	fileExtensionPair	updateCommitJaccardSimilarity	GroupRank	filePair	filePairWithRelative
34	tsx↔tsx	0.230769	5	App↔App	examples/custom-q-parsing/src/App.tsx↔examples/modal/src/App
35	tsx↔tsx	0.214286	6	App↔App	examples/auth/src/App.tsx↔examples/modal/src/App
36	tsx↔tsx	0.191275	7	index↔index	packages/react-router-native/index.tsx↔packages/react-router-native/index
37	tsx↔tsx	0.161137	8	hooks↔components	packages/react-router/lib/hooks.tsx↔packages/react-router/lib/component
38	tsx↔tsx	0.156250	9	data-browser-router-test↔index	packages/react-router-dom/_tests_/data-browser-router-test.tsx↔packages/react-router-dom/index
39	tsx↔tsx	0.144615	10	index↔components	packages/react-router-dom/index.tsx↔packages/react-router-dom/lib/component

Co-Changed Files by their commit jaccard similarity



Find pairwise changed files with many highly ranked metrics

Find those pairwise changed files that have a high rank in many metrics by calculating a combined (weighted) score based on the ranks of each metric. This is useful to identify pairs of files that score high in most metrics, which indicates a strong co-change relationship.

	fileExtensionPair	filePair	combinedMetricsScore	updateCommitCountExtensionRank	updateCommitMinConfidenceExtensionRa
0	json↔json	package↔package	26	8	
1	json↔json	package↔package	30	8	
2	json↔json	package↔package	30	8	
3	json↔json	package↔package	30	8	
4	json↔json	package↔package	30	8	
5	json↔json	package↔package	34	7	
6	json↔json	package↔package	34	7	
7	json↔json	package↔package	34	7	
8	json↔json	package↔package	36	8	
9	json↔json	package↔package	39	7	
10	md↔json	CHANGELOG↔package	29	2	
11	md↔json	README↔package	32	7	
12	md↔json	README↔package	32	7	
13	md↔json	README↔package	32	7	
14	md↔json	README↔package	32	7	
15	md↔json	README↔package	32	7	
16	md↔json	README↔package	32	7	
17	md↔json	README↔package	33	8	
18	md↔json	README↔package	34	7	
19	md↔json	README↔package	34	7	
20	md↔md	README↔README	15	10	
21	md↔md	README↔README	26	12	
22	md↔md	README↔README	26	12	
23	md↔md	README↔README	31	10	
24	md↔md	README↔README	32	12	
25	md↔md	README↔README	32	12	
26	md↔md	README↔README	35	10	
27	md↔md	README↔README	35	10	
28	md↔md	README↔README	36	12	
29	md↔md	README↔README	41	15	
30	tsx↔tsx	App↔App	28	24	
31	tsx↔tsx	exports-test↔exports-test	32	20	
32	tsx↔tsx	descendant-routes-splat-matching-test↔layout-routes-test	32	23	
33	tsx↔tsx	App↔App	33	24	
34	tsx↔tsx	App↔App	33	24	
35	tsx↔tsx	App↔App	35	24	
36	tsx↔tsx	App↔App	35	24	

	fileExtensionPair	filePair	combinedMetricsScore	updateCommitCountExtensionRank	updateCommitMinConfidenceExtensionRa
37	tsx↔tsx	App↔App	39	23	
38	tsx↔tsx	index↔index	44	1	
39	tsx↔tsx	server↔data-static-router-test	55	13	

Pairwise changed files with pareto-optimal metrics

A pair (count, confidence, jaccard, lift) is Pareto-optimal if there is no other pair that is better or equal in all metrics and strictly better in at least one. In other words, it is not "dominated" by any other pair.

The frontier = the “best tradeoffs.”

Pairwise changed files with pareto-optimal metrics - not considering file extensions

	filePair	combinedMetricsScore	updateCommitCount	updateCommitMinConfidence	updateCommitJaccardSimilarity	updateC
0	CHANGELOG↔CHANGELOG	85	229	0.835766	0.717868	
1	CHANGELOG↔CHANGELOG	89	235	0.833333	0.703593	
2	package↔package	65	376	0.833703	0.708098	
3	CHANGELOG↔package	35	180	0.772532	0.614334	
4	package↔package	26	20	0.625000	0.434783	
5	package↔package	34	21	0.600000	0.428571	
6	package↔package	34	21	0.600000	0.428571	
7	package↔package	34	21	0.600000	0.428571	
8	README↔README	15	8	0.888889	0.800000	1
9	README↔README	41	3	0.750000	0.428571	2

Pairwise changed files with pareto-optimal metrics - using ranks grouped by file extensions

	fileExtensionPair	filePair	combinedMetricsScore	updateCommitCount	updateCommitMinConfidence	updateCommitJaccardSimila
0	md↔md	use-submit↔CHANGELOG	404	3	0.073171	0.009
1	tsx↔tsx	index↔data-browser-router-test	263	6	0.063158	0.030
2	tsx↔tsx	useOutlet-test↔index	253	3	0.214286	0.011
3	md↔md	create-browser-router↔tutorial	311	3	0.053571	0.022
4	tsx↔tsx	index↔components	268	4	0.173913	0.014
5	tsx↔tsx	data-browser-router-test↔nav-link-active-test	285	3	0.068182	0.019
6	tsx↔tsx	index↔navigate-test	271	5	0.161290	0.017

WordCloud of git authors

	word	frequency
0	Michael Jackson	1832
1	Ryan Florence	1119
2	Matt Brophy	764
3	Jimmy Jia	381
4	Tim Dorr	326
5	Chance Strickland	180
6	Mateusz Zatorski	124
7	Logan McAnsh	99
8	Dan Abramov	92
9	Paul Sherman	77

Wordcloud of git authors

